A Bayesian DAG is a map I can learn from

Professor, I am using a Bayesian DAG to connect information about a student with their final course result. I find it helpful to think of this as two parts: a map, and a way to learn from data.

The DAG is the map. Each box represents something I measure, and the arrows show the relationships I assume. DAG stands for directed acyclic graph. In plain language, the arrows have a direction, and following them never brings me back to where I started.

The picture shows one path in my larger model: wellness connects to confidence, confidence connects to online activity, activity connects to assessment scores, and those scores connect to the final result. I chose these connections; the arrows do not prove that one thing causes another.

The Bayesian part describes how I learn the strengths of these relationships. I start with cautious estimates and update them using earlier students’ data.

With that idea in place, I will show you what information I put into the map.

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I fill the map with four kinds of information

To fill the map, I brought together the grade records, student surveys and course website activity. Although the files contain thousands of repeated entries, my final modeling table has 191 students across six terms, from Fall 2024 through Spring 2026.

I grouped the inputs into four areas. Wellness describes how students reported feeling. Motivation includes their confidence in learning, the value they see in the course, and their ratings of barriers and support. Online activity includes page views and recorded participation. Assessment history includes quiz, paper and other assignment scores.

I converted assignment marks into percentages so scores with different maximum points could be compared, and I removed repeated activity records. I then compared a smaller set of ten assessment measures with a larger set of sixteen.

The surveys and activity summaries use the first four assumed teaching weeks. However, I still need to confirm when each assessment score was released. That means this is a study using past records, rather than a verified week-four forecast.

Once these inputs are prepared, the next step is to learn from earlier students.

I learn from past students, then make a forecast

I keep the arrows fixed. For each box, the model learns how much weight to give the inputs pointing into it, and how much variation those inputs do not explain.

I start with cautious estimates and update them using earlier students’ inputs and known final results. To predict another student’s result, I use their available inputs while leaving their final result unknown.

The output is a predicted percentage and a range. In this historical example, the estimate was 83.1 percent and the reported range was about 77.2 to 89.0. The student’s final result was 81.0 percent, which I show here only to check the forecast afterward.

A 90 percent range is meant to include the actual result for about nine out of ten similar students. I have to check whether that happens; it is not a guarantee for one student. I adjusted these ranges using earlier prediction errors, but they still depend on simplified assumptions.

Missing surveys also need care. I used typical training values when learning, while allowing unknown inputs to remain uncertain when predicting.

Now that I have explained the forecast, I will show how I checked it.

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I learn from earlier terms and check later students

To check whether this approach can work beyond the students it learned from, I followed the order of the course terms. Within the older terms, I repeatedly learned from earlier classes and checked the next class. That helped me choose how many assessment scores to include and how cautious the model should be.

After making those choices, I used the selected model to learn from 148 students from the first five terms. I then predicted the final percentages of 43 students from Spring 2026, without using their final results to fit that model.

The timeline shows that separation. I also kept the data preparation within the training group, so the later students did not determine the values used to prepare the earlier data.

There is an important limit to this check. Spring 2026 has already been examined during earlier versions of this project. It gives me useful evidence, but I still need a new class that I have not inspected before.

With that testing process explained, I can now show the results and what they do and do not tell me.

The extra inputs have not improved average error yet

The first result is that the forecasts are fairly close on average in this historical check. The average distance between the full model’s prediction and the recorded final percentage was 2.83 percentage points. This is an average error, not a percentage accuracy, and some students had much larger errors.

The comparison tells me something more specific. A simple forecast using the earlier students’ average final grade had an average error of 8.13 points. Using the available assessment scores reduced that error to 2.82. Adding wellness, motivation and online activity gave 2.83, which is almost identical and slightly worse on this measure.

So I have not shown that those additional inputs improve the final-grade prediction. I also cannot conclude that they will never help. The larger assessment set already contains a lot of information about the final result. Some assessment scores are also part of the final grade, so they naturally carry information about it. The survey data are incomplete as well.

The reported ranges included the actual result for 39 of the 43 students, but the largest prediction error was about 25.7 points. Those findings are why I need to examine both the predictions and their uncertainty.

My next step is to test whether the additional inputs help earlier in the course.

Next, I will test the extra inputs earlier

My next experiment will focus on a time when the information could still be useful for student support. I would first choose a prediction date, such as the end of week four, and verify which surveys, activity records and assessment scores were actually available by that date.

I would then predict the same students using the same available grades. I would add wellness alone, motivation alone, and online activity alone, and finally compare a model that includes all of them. I would keep the rest of the comparison consistent so an improvement cannot simply come from using later grades or a different set of students.

I would repeat this process across later terms and check both the average error and the prediction ranges. I would also investigate the 78 students who are missing early survey information, because that may affect the comparison.

Before examining a new class, I would agree on what size of improvement would be useful and keep the chosen model unchanged. If the additional inputs do not help, that is still a valid result.

I would like your feedback on the prediction date and what evidence would make this useful for early student support.